ANALYSIS OF SWITCH DIVERSITY WITH POST-EXAM SELECTION IN GENERALIZED CORRELATED FAIDNG ENVIROMENT

by

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ABSTRACT

Diversity is a commonly used technique in wireless system to combat channel fading. The idea behind diversity is to send the same data over independent fading paths and combine the received signals in a way such that the amount of fading experienced by the resultant signal is reduced. There are several combining schemes which vary in complexity and overall performance. Low complexity combining scheme such as switched combining is attractive from the receiver simplicity perspective. In this report, we study a particular switched combining scheme namely switch and examining combining with post examining selection (SECps).This new scheme improves the performance of conventional switch combining, whereas requiring less path estimation and path switching than selection combining (SC). While conventional switched combining always switch to other paths if unacceptable current path is found even when the switch-to path has a lower SNR, SECps allows the receiver to use the strongest path when no acceptable path is found after examining all diversity paths.

In this report, we focus on the study and analysis of SECps performance over correlated Nakagaimi-*m* fading. We derive the statistics of the combined output signal to noise ratio (SNR) of SECps, which are then applied to the performance and complexity analysis of SECps. In particular, we point out the effect of fading correlation and fading severity on system performance and operation complexity. From the selected numerical results, we illustrate that fading correlation induces non-negligible degradation on SECps system performance.

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To My Dear Family

Chapter 1

Introduction

Wireless communication is by any measure the fastest growing segment of the communication industry [2]. The explosive growth of wireless less system such as cellular system have experienced exponential growth over the last decade and there are currently around two billion user worldwide. In addition, wireless local area networks currently supplement or replace wired networks in many homes, businesses and campuses. Design of wireless networks differs fundamentally from wired network design due to the nature of wireless channel. This channel is an unpredictable and difficult communications medium. As a signal propagate through a wireless channel, it experiences random fluctuation in time due to the reflected, scattered, diffracted, or refracted by the various obstacles in the channel. This aspect of the transmission fluctuation, where the signal loses its deterministic nature and become random in time and space is described in term of fading. The characteristics of fading make it difficult to design reliable system to guarantee performance. Thus, fading is a significant challenge key in wireless network design. In this report we will focus mainly on the study of diversity technique which is one of the efficient ways to combat fading.

In this chapter, we present an overview of diversity technique and briefly provide the detail of various types of diversity combiners. In particular, we introduce the switch and examine combining with post-examining selection scheme (SECps) which will be

discussed in more detail in the remainder chapters. Then we present the contribution which we obtain from the study of SECps performance. Finally, the organization of this report is given as a guideline.

1.1 Diversity Technique

One of the most powerful yet simplest techniques to overcome the destructive effects of fading in wireless communication system is diversity. Diversity-combining consists of receiving redundantly the same information-bearing signal over two or more fading channels, then combining these multiple replicas at the receiver in order to increase the overall received SNR. The intuition behind this concept is to exploit the low probability of concurrence of deep fades in all diversity channels to lower the probability of error and of outage [1].

Diversity is an efficient way to combat channel fading due to following facts [3]. Firstly, the degradation of transmission quality due to channel fading cannot be simply overcome by increasing the transmitted signal power. This is because, even with high transmitted power, when the channel is in deep fading, the instantaneously received SNR can still be very low, resulting in high probability of transmission error due to deep fading period. Secondly, in wireless communications, the power available on the reverse link is severely limited by the battery capacity in hand-held subscriber units. With diversity, the required transmitted power can be greatly reduced. Thirdly, cellular communications system are mostly interference limited and once again, mitigation of channel fading by diversity reception can translate into improved interference tolerance which in turn, means greater ability to support additional users and therefore higher system capacity.

In diversity techniques, there are various ways to obtain the multiple replicas by extracting the signals from different radio paths [1]. For example, from space by using multiple-transmitting or multiple-receiving antennas, which is called space diversity,

from frequency by using multiple-frequency channels separated by at least the coherent bandwidth of the channel, which is called frequency diversity, from time by using multiple time slots separated by at least coherent time of the channel, which is called time diversity, or from multipath by resolving multipath components at different delays, which is called path diversity.

After receiving the multiple received signals from different paths, the receiver diversity combines the received signals to obtain a resultant signal which is then pass through a standard demodulator. The combining can be done in several ways which vary in complexity and overall performance. In the following section, we will describe the different combining techniques and their performance. These techniques entail various tradeoffs between performance and complexity.

1.2 Diversity-Combining Schemes

There are four principle types of combining technique which depend essentially on the (1) complexity restriction put on the communication system and (2) amount of channel state information (CSI) available at the receiver [1].

Maximal-Ratio Combining (MRC) is the optimal combining scheme, which can provide the best performance among other schemes. As such, the MRC scheme also has high implementation complexity and requires knowledge of all channel fading parameters. This combining technique assumes that the receiver is able to accurately estimate the amplitude fading and carrier phase distortion for each diversity path. With knowledge of the complex channel gain the receiver coherently demodulates the receiver signal from each path. The coherently detected signal is then weighted by the corresponding amplitude gain. Then, the weighted received signals from all branches are sum together and applied to the decision device [4].

Another alternative approach, which reduces the complexity of MRC, is the Equal gain combining scheme (EGC). This combining technique weights all the signals equally after coherent detection. Then, the coherently detected signals from all branches are simply added and applied to the decision device. As the receiver does not need to estimate the amplitude fading, it complexity is reduced.

The two former combining schemes (MRC and EGC) require processing received signals from all diversity paths. On the other hand, Selection Combining scheme (SC) only process one of the diversity path, thus the coherent sum of the individual path is not required. With SC, the receiver monitors the signal to noise ratio (SNR) value of each diversity channel and chooses the strongest diversity path, which is the one with the highest SNR for data reception. Therefore, compared with the preceding two schemes, SC has much lower complexity.

Switched combining is an attempt at further simplifying the complexity of selection combining with the expense of the loss in performance. In this case, rather than continually connecting the diversity path with the best quality, the receiver selects a particular path until its quality drops below a predetermined threshold. When this happens, the receiver switches to another diversity path. This result in a complexity reduction relative to selection combining in that the simultaneous and continuous monitoring of all diversity paths is no longer necessary [1].

1.3 Variants of Switched Combining

Switch and stay combining (SSC) is obviously the least complex diversity scheme since it only monitors on the current path and switches to the other path whenever the SNR of the current path falls below the predetermined threshold regardless of the switch-to path's quality. This mode of operation is based the concept of the low probability of concurrence of deep fade in all paths, hence, the switch-to path has a good chance of being less severely faded. However, it might so happen that the switch-to path has even

lower SNR than the current path. In this case, path switching actually degrades the system performance. Since the SSC does not select the branch with highest SNR, its performance is between that of no diversity and ideal SC.

When the number of available diversity paths is greater than two, an alternative switched combining scheme, named switch and examine combining (SEC) should be used to take advantage of additional paths. This switched combining scheme tries to use an acceptable path by examining as many paths as necessary. The receiver with SEC monitors the current path and when its SNR falls below the threshold, the receiver examines the other path until it finds an acceptable path and switches to that new acceptable path. Nevertheless, when no acceptable path is found, the receiver with SEC randomly chooses an unacceptable path as a data reception.

Since all available diversity paths are examined anyway in the case of SEC, a more preferable scheme to achieve better performance is to select the strongest one among unacceptable paths. The modified switched combining scheme, which allows the receiver with SEC and SSC to use the strongest path when no acceptable path is found, is named switch and examine combining with post-examining selection (SECps). In this combining scheme, the receiver operates in similar way as SEC but when the receiver can not find the acceptable path, it chooses the path with the highest SNR rather than randomly select path like conventional SEC.

1.4 Contribution

The previous work on SECps in [5] studied the performance of SECps over independent identically distributed (i.i.d.) fading channels. However, independent fading occurs only if antenna elements are placed sufficiently apart, which is not always realizable in practice due to, for example, the size limitation of mobile terminals. In this kind of terminal, the fading among the channel is correlated, resulting in degradation of the

diversity gain [5]. Therefore, it is important to understand how the correlation between received signals affects the system performance.

In this report, we focus on the analysis of SECps performance over correlated Nakagami-*m* fading channel. In particular, we present the SECps's improvement of performance over conventional switched combining scheme. In additional, simulations are performed to check the accuracy of the purposed mathematical analysis. Then, the effect of fading correlation as well as the fading severity on system performance is pointed out. Some specific contributions of the report are as follows:

1) Derivation of statistics of SECps combined output SNR over correlated Nakagami-*m* fading channel
2) Analysis of the effect of correlated fading and fading severity on SECps performance criteria: outage probability and average error rate, over Nakagami-*m* fading channel
3) Analysis of the effect of correlated fading and fading severity channel on SECps complexity issues: average number of path estimation and switching probability, over Nakagami- *m* fading channel

1.5 Organization of the report

The remainder of this report is organized as follow. Chapter 2 contains system structure and the characteristic of mode of operation of dual-branch SECps in addition to the channel model under consideration. In chapter 3, we introduce the derivation of statistic of combined output SNR and the important performance criteria which are outage probability and average error rate on dual-branch SECps. Chapter 4 presents the analysis of complexity issues which are channel estimation and switching probability on dual-branch SECps. Simulations and numerical results are provided in chapter 3 and 4 to illustrate the proposed mathematical analysis. Finally, in chapter 5 we conclude a summary of the main results and gives suggestion for future research.

Chapter 2

System and Channel Model

In this chapter, we present the system and channel model under consideration for dual-branch SECps combiner. After introducing the receiver structure, we explain the mode of operation of SECps and describe the fading channel model used in this report.

2.1 System Structure

Figure 2.1 presents the structure of dual-branch SECps combiner. The signal replicas received over two diversity paths with instantaneous SNR denoted by $\gamma_i, i = 1, 2$ are fed to the SECps combiner. Then the SECps combiner selects the acceptable path and feeds the selected signal to the receiver for data reception. In particular, the SECps combiner estimates the SNR value of the current path and compares it with the predetermined switching threshold. If the SNR value of the current path is higher than or equal to the switching threshold, the receiver continues staying with the current path. Otherwise, the combiner searches for an acceptable path by repeating the whole process with the other path. If acceptable path cannot be found, the combiner chooses the best path with the maximum SNR value. Therefore the combined output SNR can be expressed as

$$\gamma_\Sigma = \gamma_i, \gamma_i \geq \gamma_T \text{ or } \max\{\gamma_1, \gamma_2\}, \gamma_i < \gamma_T \tag{2.1}$$

where γ_i is the SNR value of the current path and i={1,2}

From (2.1), we can see that the SECps receiver performs as the conventional SSC receiver when the SNR of the current path is higher or equal to the switching threshold. On the other hand, the SECps receiver performs as the SC receiver when the SNR of the current path falls below the switching threshold. In the next section, we will present the path updating process of SECps combiner in more detail.

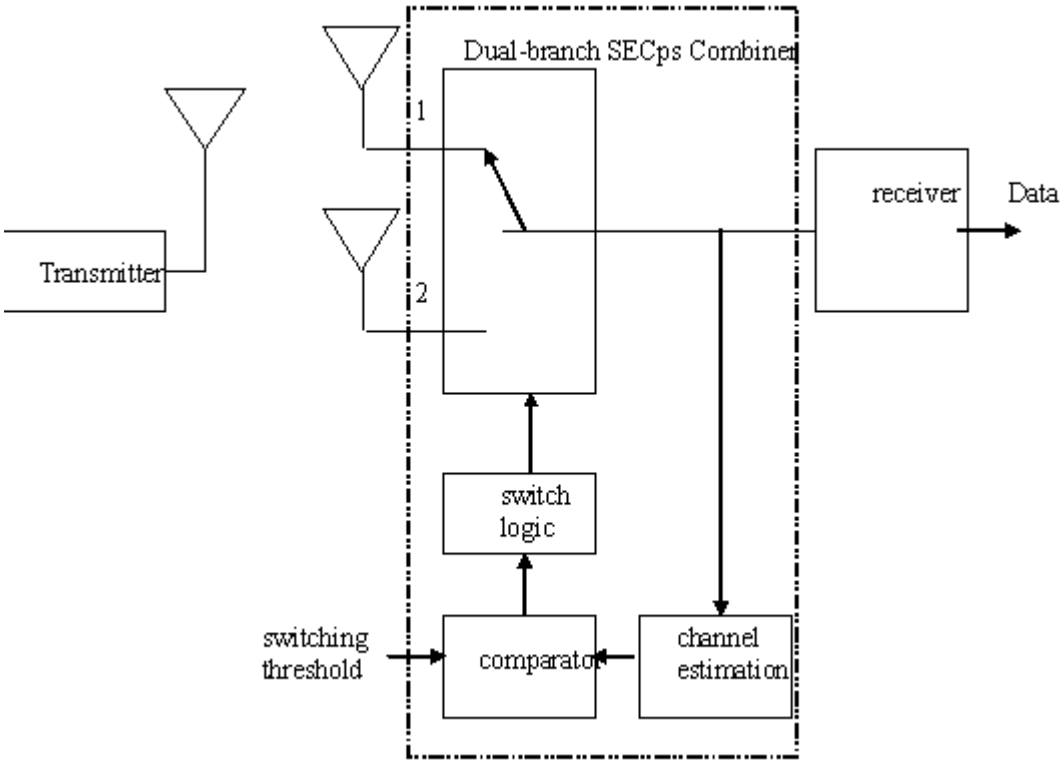


Figure 2.1 block diagram of dual-branch SECps combiner

2.2 Mode of Operation of SECps

In the operation mode of dual-branch SECps, the receiver with SECps tries to use an acceptable path which has its SNR above the predetermined threshold by examining many paths only if necessary as the case of conventional SSC. However, when an acceptable path cannot be found after examining all paths, the receiver with SECps

selects the best path with the highest SNR instead of randomly selects path as the case of conventional SSC. More specifically, the receiver first estimates the SNR of the current path denoted by γ_i and compares with the predetermined threshold denoted by γ_T. If the SNR of the current path is higher or equal to the threshold $(\gamma_i \geq \gamma_T)$, the receiver continues uses that current path. On the other, if the SNR of the current path falls below the threshold $(\gamma_i < \gamma_T)$, the receiver with SECps exams the other path. For example, if we assume the first path is the current path with its SNR lower than the threshold $(\gamma_1 < \gamma_T)$, then the SNR on the second path, denoted by γ_2, is estimated and compared with the threshold. If the SNR of the second path is higher than the threshold or the SNR of the first path $(\gamma_2 \geq \gamma_T$ or $\gamma_2 > \gamma_1)$, then the receiver switches to the second path and use this path for data reception, otherwise the receiver continues use the first path. In the later case, we can see that when acceptable path cannot be found after the receiver estimates all paths, the receiver chooses the best path with the highest SNR for data reception.

Compared to the conventional SSC, SECps requires the additional complexity for the occasional selection of the best path among all unacceptable paths. However, SECps takes full advantage of the available path estimation. In particular, by using the best unacceptable path instead of a randomly chosen on (usually the last one examined) as in SSC. Furthermore, path switching with SECps occurs only when the currently used path is unacceptable and another path is better than the current one. As such, SECps needs less path switching than SSC.

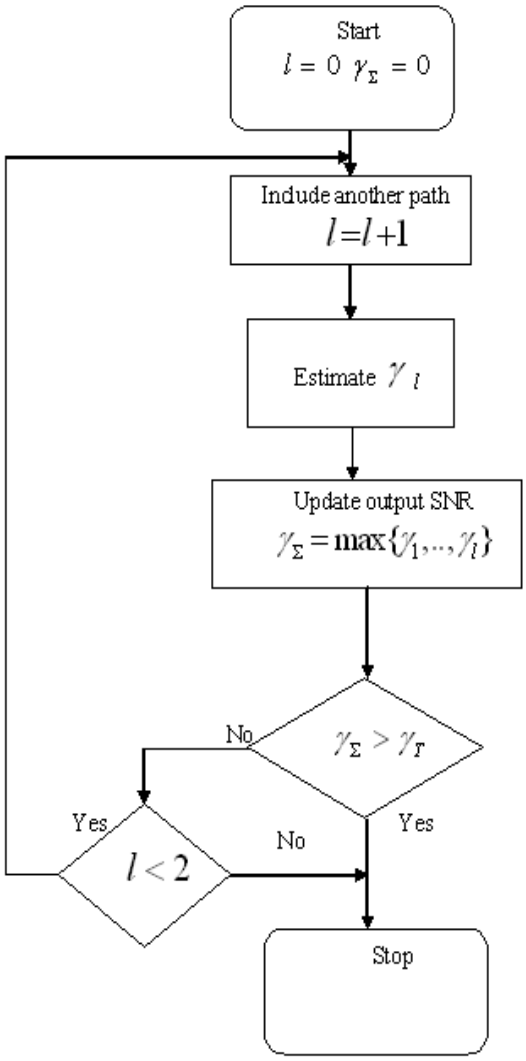


Figure 2.2 Mode of operation of dual-branch SECps [5]

2.2 Correlated Nakagami-*m* Fading Channel Model

We focus in this report the Nakagami-*m* fading environment. More specifically, the signal is assumed to propagate over correlated and identically distributed Nakagami-*m* fading channel. The fading *m* parameter is a measure of the severity of the fading: a small *m* implies severe fading, a large *m* implies more mild fading. For example, for the value of $m = 1$, Nakagami distribution becomes Rayleigh, for value of $m > 1$, Nakagami

distribution is approximately Rician, and for $m \to \infty$, Nakagami distribution becomes a non fading AWGN channel . The Nakagami-*m* probability density function is thus general enough to encompass both the Rayleigh and Rician distribution [3]. The PDF of Nakagami-*m* is given by [1,eq(2.20)]

$$p_{\alpha}(\alpha) = \frac{2m^{m}\alpha^{2m-1}}{\Omega^{m}\Gamma(m)} \exp\left(-\frac{m\alpha^{2}}{\Omega}\right), \quad \alpha \geq 0 \tag{2.2}$$

In Nakagami-*m* fading environment, the SNR per symbol γ is distributed according to a gamma distribution given by

$$p_{\gamma}(\gamma) = \frac{m^{m}\gamma^{m-1}}{\bar{\gamma}^{m}\Gamma(m)} \exp\left(-\frac{m\gamma}{\bar{\gamma}}\right), \qquad \gamma \geq 0 \tag{2.3}$$

where $\bar{\gamma}$ is an average SNR

(Note: identically distributed means all channels are assumed to have the same average SNR per symbol $\bar{\gamma}$ and the same fading parameter *m*)

In additional, it is assumed that fading is slowly varying, and hence, perfect channel estimation can be obtained. The joint PDF of the two correlated Nakagami-*m* instantaneous SNR per symbol γ_1 and γ_2 is given by [6, eq.(68)]

$$\begin{aligned} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2) = &\left(\frac{m}{\bar{\gamma}}\right)^{m+1} \frac{\gamma_1^{(m-1)/2}\gamma_2^{(m-1)/2}}{\Gamma(m)\rho^{(m-1)/2}(1-\rho)} e^{-(m/\bar{\gamma}(1-\rho))(\gamma_1+\gamma_1)} \\ &\times I_{m-1}\left(\frac{2m\sqrt{\rho}}{(1-\rho)\bar{\gamma}}\sqrt{\gamma_1\gamma_2}\right), \qquad \gamma_1 \geq 0, \gamma_2 \geq 0 \end{aligned} \tag{2.4}$$

with $\Gamma(.)$ being the Gamma function, $I_v(.)$ being the first kind and *v*th-order modified Bessel function, and ρ is the power correlation coefficient between the two diversity paths defined as

$$\rho = \frac{\mathrm{cov}(a_1^2, a_2^2)}{\sqrt{\mathrm{var}(a_1^2)\,\mathrm{var}(a_2^2)}}, \qquad 0 \le \rho < 1 \tag{2.5}$$

where a_1, a_2 being the correlated Nakagami-*m* envelops of the input signal.

Chapter 3

Performance Analysis of Dual-Branch SECps

In this chapter, we study the performance of dual-branch SECps over correlated Nakagami-*m* fading channel. After deriving the statistics of SECps, we then apply them to the outage probability and average error rate analysis. In additional, simulations and numerical results are provided to illustrate the accuracy of the purposed mathematic.

3.1 Statistics of Combined Output SNR

In this section, we derive the statistical characterization of the combined output SNR with dual-branch SECps in the term of the cumulative distribution function (CDF) and probability density function (PDF). The PDF of switch combining (SC) and switch and stay combining (SSC) are also given for comparison.

3.1.1 Statistic of dual-branch SECps

In dual-branch SECps, the combiner selects one of the two diversity paths for data reception. Thus, the combined output SNR is equal to the SNR on one path. In order to write the CDF of the combined output SNR, we consider the following two scenarios.

(i) The SNR of the current path is equal to or above the threshold $(\gamma_1 \geq \gamma_T)$. Thus, the output SNR is equal to that of the current path. $(\gamma_\Sigma = \gamma_1)$.

(ii) The SNR of the current path is below the threshold $(\gamma_1 < \gamma_T)$. Thus, the output SNR is equal to the maximum SNR of both paths. $(\gamma_\Sigma = \max\{\gamma_1, \gamma_2\})$.

Note: assuming the first path is used as the current path

Therefore, we can write the CDF of the combined output SNR as follow.

$$\begin{aligned} P_{\gamma_\Sigma}(x) &= \Pr[\gamma_\Sigma < x] \\ &= \Pr[\gamma_\Sigma = \gamma_1 < x, \gamma_1 > \gamma_T] + \Pr[\gamma_\Sigma = \max[\gamma_1, \gamma_2] < x, \gamma_1 < \gamma_T] \end{aligned} \tag{3.1}$$

where γ_1, γ_2 is the SNR of the first and the second branch respectively, and γ_T is the switching threshold level.

By considering two scenarios of the value of x: $x \geq \gamma_T$ and $x < \gamma_T$, we can rewrite the CDF of combined output SNR in (3.1) as follow.

$$P_{\gamma_\Sigma}(x) = \begin{cases} \Pr[\gamma_T < \gamma_1 < x] + \Pr[\gamma_1 < \gamma_T, \gamma_2 < x], & x \geq \gamma_T \\ \Pr[\gamma_1 < x, \gamma_2 < x], & x < \gamma_T \end{cases} \tag{3.2}$$

For dual-branch diversity with correlated branch statistics, we can express (3.2) in the terms of common CDF $P_{\gamma_\Sigma}(\gamma) = p(\gamma_\Sigma \leq \gamma)$ and p_{γ_i} where $i = 1, 2$ as

$$P_{\gamma_\Sigma}(\gamma) = \begin{cases} \int_{\gamma_T}^{\gamma} p_{\gamma_1}(\gamma_1) d\gamma_1 + \int_0^{\gamma_T} \int_0^{\gamma} p_{\gamma_1,\gamma_2}(\gamma_1, \gamma_2) d\gamma_2 d\gamma_1 & \gamma \geq \gamma_T \\ \int_0^{\gamma} \int_0^{\gamma} p_{\gamma_1,\gamma_2}(\gamma_1, \gamma_2) d\gamma_2 d\gamma_1 & \gamma < \gamma_T \end{cases} \tag{3.3}$$

We can notice that, the first term in (3.3) is equal to the CDF of a single branch receiver. Thus, with correlated Nakagami-*m* fading environment, this first term can be expressed as

$$\int p_{\gamma_1}(\gamma_1)d\gamma_1 = 1 - \frac{\Gamma(m, \frac{m}{\bar{\gamma}}\gamma)}{\Gamma(m)} \tag{3.4}$$

The integral of joint PDF from $\gamma = 0$ to γ_T on the second term in (3.3) can be expressed in the term of $A(\gamma)$ from [4,eq.(70)] as

$$A(\gamma) = \int_0^{\gamma_T} p_{\gamma_1,\gamma_2}(\gamma, \gamma_2)d\gamma_2 \tag{3.5}$$

where from [6,eq.(73)]

$$A(\gamma) = \left(\frac{m}{\bar{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\bar{\gamma})}}{\Gamma(m)} \times \left[1 - Q_m\left(\sqrt{\frac{2m\rho\gamma}{(1-\rho)\bar{\gamma}}}, \sqrt{\frac{2m\gamma_T}{(1-\rho)\bar{\gamma}}}\right)\right] \tag{3.6}$$

Thus, the second term in (3.3) can be expressed as

$$\int_0^{\gamma_T}\int_0^{\gamma} p_{\gamma_1,\gamma_2}(\gamma_1, \gamma_2)d\gamma_2 d\gamma_1 = \int_0^{\gamma} A(\gamma)d\gamma_1 \tag{3.7}$$

By substituting (3.4) into the first term, $A(\gamma)$ from (3.6) into the second term, and the joint PDF of the instantaneous SNR of γ_1 and γ_2 from (2.4) into the third term of (3.3), the CDF of combined output SNR (γ_Σ) can be rewritten as

$$P_{\gamma_\Sigma}(\gamma) = \begin{cases} 1 - \dfrac{\Gamma(m, \frac{m}{\bar{\gamma}}\gamma)}{\Gamma(m)} + \displaystyle\int_0^{\gamma} \left(\frac{m}{\bar{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\bar{\gamma})}}{\Gamma(m)} \times \left[1 - Q_m\left(\sqrt{\frac{2m\rho\gamma}{(1-\rho)\bar{\gamma}}}, \sqrt{\frac{2m\gamma_T}{(1-\rho)\bar{\gamma}}}\right)\right] d\gamma_1 \ , \ \gamma \geq \gamma_T \\ \displaystyle\int_0^{\gamma}\int_0^{\gamma} \left(\frac{m}{\bar{\gamma}}\right)^{m+1} \frac{\gamma_1^{(m-1)/2}\gamma_2^{(m-1)/2}}{\Gamma(m)\rho^{(m-1)/2}(1-\rho)} e^{-(m/\bar{\gamma}(1-\rho))(\gamma_1+\gamma_1)} \times I_{m-1}\left(\frac{2m\sqrt{\rho}}{(1-\rho)\bar{\gamma}}\sqrt{\gamma_1\gamma_2}\right) d\gamma_1\gamma_2 \ , \ \gamma < \gamma_T \end{cases} \tag{3.8}$$

Differentiating $P_{\gamma_\Sigma}(\gamma)$ in (3.8) with respect to γ, we obtain a formula for the PDF of the combined output SNR, $p_{\gamma_\Sigma}(\gamma)$ as

$$p_{\gamma_\Sigma}(\gamma)=\begin{cases}\left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)} + A(\gamma) & \gamma \geq \gamma_{\mathrm{T}} \\ 2\times\left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)}\left[1-Q_m(\sqrt{\frac{2\rho\gamma m}{\overline{\gamma}(1-\rho)}},\sqrt{\frac{2\gamma m}{\overline{\gamma}(1-\rho)}})\right] & \gamma < \gamma_{\mathrm{T}}\end{cases} \tag{3.9}$$

In the next subsection, we provide the PDF of SSC and SC in order to compare with the PDF of SECps.

3.1.2 Comparison with the statistics of dual-branch SSC and dual-branch SC

Under the condition of correlated Nakagami-*m* fading distribution, Abu-Dayya and Beaulieu have shown in [6,eq(69)] that the PDF of dual-branch SSC is given as

$$p_{\gamma_{SSC}}(\gamma)=\begin{cases}A(\gamma) & \gamma < \gamma_T \\ \left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)} + A(\gamma) & \gamma \geq \gamma_T\end{cases} \tag{3.10}$$

Where the PDF of the output SNR of a dual-branch SC over correlated Nakagami-*m* fading is given by [1, eq.(9.235)]

$$\begin{aligned}p_{\gamma_{SC}(\gamma)} = {} & \frac{(m/\overline{\gamma}_1)^m \gamma^{m-1}}{\Gamma(m)}\exp\left(\frac{-m\gamma}{\overline{\gamma}_1}\right)\left[1-Q_m(A_1\sqrt{2\rho\gamma},A_2\sqrt{2\gamma})\right] \\ & + \frac{(m/\overline{\gamma}_2)^m \gamma^{m-1}}{\Gamma(m)}\exp\left(\frac{-m\gamma}{\overline{\gamma}_2}\right)\left[1-Q_m(A_2\sqrt{2\rho\gamma},A_1\sqrt{2\gamma})\right]\end{aligned} \tag{3.11}$$

where A_l is given by $A_l=\sqrt{m/[\overline{\gamma}_l(1-\rho)]}$

When assuming that the average SNR in all branch is equal ($\bar{\gamma}_1 = \bar{\gamma}_2 = \bar{\gamma}$), we can write the PDF from (3.11) as

$$p_{\gamma_{SC}(\gamma)} = 2 \times \frac{(m/\bar{\gamma})^m \gamma^{m-1}}{\Gamma(m)} \exp\left(\frac{-m\gamma}{\bar{\gamma}}\right)\left[1 - Q_m\left(\sqrt{\frac{2\rho\gamma m}{\bar{\gamma}(1-\rho)}}, \sqrt{\frac{2\gamma m}{\bar{\gamma}(1-\rho)}}\right)\right] \quad (3.12)$$

By comparing the PDF of SECps from (3.9) to that of SSC in (3.10) and SC in (3.12) respectively, we can rewrite the PDF of SECps as follow.

$$p_{\gamma_{SECps}}(\gamma) = \begin{cases} p_{\gamma_{SSC}}(\gamma) & \gamma \geq \gamma_T \\ p_{\gamma_{SC}}(\gamma) & \gamma < \gamma_T \end{cases} \quad (3.13)$$

From (3.13), we can see that the PDF of SECps is equal to that of SSC when its simultaneous SNR, γ is greater than or equal to the switching threshold $(\gamma \geq \gamma_T)$. On the other hand, when its SNR becomes less than the switching threshold $(\gamma < \gamma_T)$, the PDF of SECps becomes the same as that of SC.

3.2 Outage Probability

One of the standard performance criterion characteristics of diversity system operating over fading channel is the so-called outage probability denoted by P_{out} and defined as the probability that the instantaneous error probability exceeds a specified value that the output SNR, γ, falls below a certain specified threshold, γ_{th}. Mathematically speaking, we have [1]

$$P_{out} = \int_0^{\gamma_{th}} p_\gamma(\gamma) d\gamma \quad (3.14)$$

which is the cumulative distribution function (CDF) of γ, namely $P_\gamma(\gamma)$,evaluated at $\gamma = \gamma_{th}$.

In this section, we analyze the outage probability of dual-branch SECps over correlated Nakagami-*m* fading and noise-limited environment. The outage probability is calculated by evaluating the CDF of the combined SNR at the outage threshold level (γ_{th}) as follow.

$$P_{out} = \Pr[\gamma_{\Sigma} < \gamma_{th}] = P_{\gamma_{\Sigma}}(\gamma_{th}) \tag{3.15}$$

We can calculate the outage probability of SECps by substituting (3.15) into the CDF form of SECps in (3.8) as

$$P_{out} = P_{\gamma_{\Sigma}}(\gamma_{th}) = \begin{cases} 1 - \dfrac{\Gamma(m, \dfrac{m}{\overline{\gamma}}\gamma_{th})}{\Gamma(m)} + \displaystyle\int_0^{\gamma_{th}} \left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1} e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)} \times \left[1 - Q_m\left(\sqrt{\frac{2m\rho\gamma}{(1-\rho)\overline{\gamma}}}, \sqrt{\frac{2m\gamma_T}{(1-\rho)\overline{\gamma}}}\right)\right] d\gamma_1, \ \gamma_{th} \geq \gamma_T \\ \displaystyle\int_0^{\gamma_{th}}\int_0^{\gamma_{th}} \left(\frac{m}{\overline{\gamma}}\right)^{m+1} \frac{\gamma_1^{(m-1)/2}\gamma_2^{(m-1)/2}}{\Gamma(m)\rho^{(m-1)/2}(1-\rho)} e^{-(m/\overline{\gamma}(1-\rho))(\gamma_1+\gamma_1)} \times I_{m-1}\left(\frac{2m\sqrt{\rho}}{(1-\rho)\overline{\gamma}}\sqrt{\gamma_1\gamma_2}\right) d\gamma_1\gamma_2 \ , \ \gamma_{th} < \gamma_T \end{cases} \tag{3.16}$$

The integral equation in (3.16) is then used to perform the simulation of outage probability performance in the next subsection.

3.2.1 Simulation and Numerical Results

Figure 3.1 shows the outage probability of dual-branch SECps, dual-branch SSC and dual-branch SC as the function of normalized outage threshold over correlate Nakagami-*m* fading environment. While comparing the outage performance of SECps with that of SSC and SC, we can see that SECps has the same outage probability performance as SC when the outage threshold is less than the switching threshold ($\gamma_{th} < \gamma_T$) and as conventional SSC when the outage threshold is higher than the switching threshold ($\gamma_{th} > \gamma_T$). This can be explained as follow. When $\gamma_{th} < \gamma_T$, the receiver with SECps experiences outage only when the maximum SNR of all diversity paths fall below the

outage threshold (γ_{th}) which is the same as the case of the receiver with SC. On the other hand, when $\gamma_{th} > \gamma_T$, the receiver with SECps experiences outage when the SNR of the first acceptable path falls below the outage threshold (γ_{th}), or when the maximum SNR of all diversity paths is smaller than the switching threshold (γ_T), which is equivalent to the outage performance of conventional SSC.

In figure 3.2, we study the effect of fading correlation on the outage performance of SECps over Nakagami-*m* fading channel. We use various values of correlation coefficient, i.e. $\rho = 0.1, 0.5$ and 0.9 and compare the outage performance over different Nakagami-*m* fading parameters, i.e. *m*=1,3. From figure 3.2, we can see that with fixed normalized outage threshold, the outage probability increases as the fading correlation increases. This is as expected since diversity technique is based on the concept that signals traveling through independent paths are most likely not to experience deep fade simultaneously. Therefore, when the diversity paths are correlated, diversity technique cannot achieve its optimum performance. When all paths are independent, i.e. $\rho = 0$, the system can achieve the lowest outage probability as we can see from figure 3.2.

In additional, it can be seen from figure 3.2 that the effect of fading correlation is much more significant in a lower amount of fading (higher *m* parameter) and this effect becomes less significant when $\gamma_{th} > \gamma_T$.

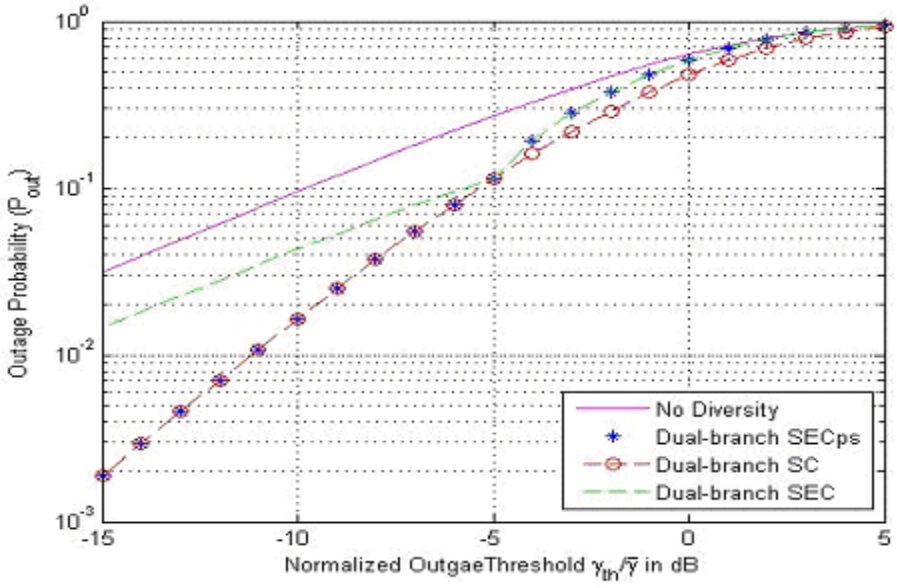


Figure 3.1 Outage probability of dual-branch SECps as function normalized outage threshold $(\gamma_{th}/\bar{\gamma})$ over correlated Nakagami-*m* fading channel in comparison with SC and SSC $(\rho = 0.5, m = 1$ and $\gamma_T/\bar{\gamma} = -5$ dB)

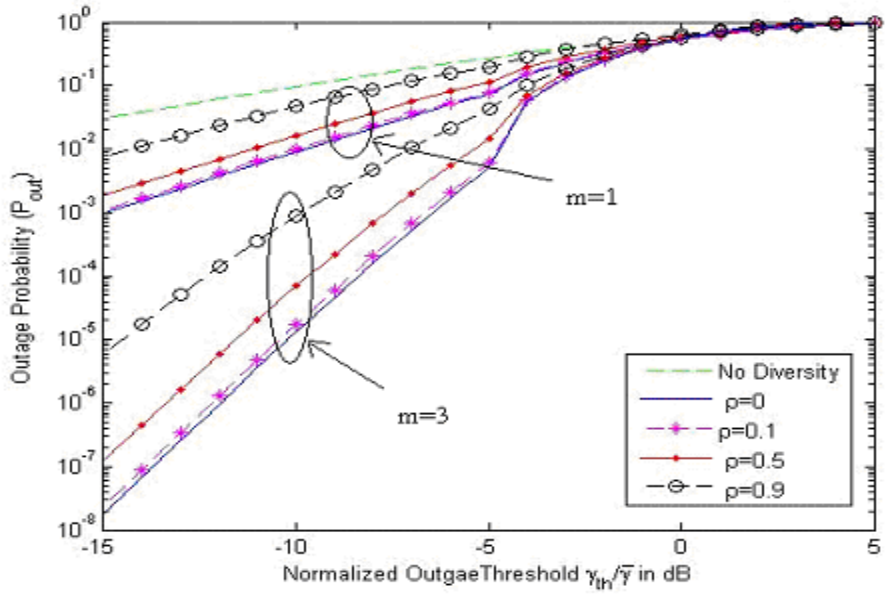


Figure 3.2 Outage probability of dual-branch SECps as function of normalized outage threshold $(\gamma_{th}/\bar{\gamma})$ over correlated Nakagami-*m* fading channel with various values of correlation coefficient, ρ at different Nakagami parameters, *m*

3.3 Average Error Rate

Another standard performance criterion of diversity system and the most revealing about the nature of the system behavior is average bit error probability. In AWGN channel the probability of symbol error, P_S depends on the received SNR, γ_S. In a fading environment, the received signal power varies randomly over distance or time due to shadowing and/or multipath fading. Thus, in fading, γ_S is a random variable with distribution $p_{\gamma_s}(\gamma)$, and therefore, symbol error probability is also random. The average error probability of symbol is computed by integrating the error probability in AWGN over the fading distribution [2].

$$\overline{P}_S = \int_0^\infty P_S(\gamma) p_\gamma(\gamma) d\gamma \tag{3.17}$$

In this section, since only the received signal power in each diversity path is required for the operation of SECps, we consider non-coherent or differentially coherent modulation schemes. We can calculate the average error rate of different modulation schemes with SECps by averaging the conditional error rate over the PDF of γ_Σ as

$$\overline{P}_S = \int_0^\infty P_S(\gamma) p_{\gamma_\Sigma}(\gamma) d\gamma \tag{3.18}$$

where $P_S(\gamma)$ denotes the probability of symbol error in AWGN channel with SNR, γ and $p_{\gamma_\Sigma}(\gamma)$ denotes the PDF of SECps combining receiver.

As an example, we consider the average error rate of a binary differential phase shift keying modulation (BDPSK) with dual-branch SECps over identically correlated Nakagami-*m* fading. By substituting $p_{\gamma_\Sigma}(\gamma)$ in (3.9) and $P_S(\gamma)$ of BDPSK $= 0.5\exp(-\gamma)$ to (3.18), we have

$$\bar{P}_S = \begin{cases} \int_{\gamma_T}^{\infty} 0.5\exp(-\gamma)\left[\left(\frac{m}{\bar{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\bar{\gamma})}}{\Gamma(m)} + A(\gamma)\right] d\gamma & \gamma \geq \gamma_T \\ \int_{0}^{\gamma_T} 0.5\exp(-\gamma)\left(2\times\left(\frac{m}{\bar{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\bar{\gamma})}}{\Gamma(m)}\left[1-Q_m(\sqrt{\frac{2\rho\gamma m}{\bar{\gamma}(1-\rho)}},\sqrt{\frac{2\gamma m}{\bar{\gamma}(1-\rho)}})\right]\right) d\gamma & \gamma < \gamma_T \end{cases} \quad (3.19)$$

The integral equation in (3.19) is then used to perform the simulation of the average error rate of SECps in the next subsection.

3.2.1 Simulation and Numerical Results

Figure 3.3 depicts the average error rate of BDPSK with dual-branch SECps as a function of average SNR per path $(\bar{\gamma})$. The switching threshold is fixed to 8 dB. As we can see, in the low SNR region, dual-branch SECps has the same performance as dual-branch SC, and when SNR increase, the performance of SECps degrades and eventually becomes the same as that of dual-branch SSC. These can be explained as when the average SNR per path $(\bar{\gamma})$ is small in comparison to the switching threshold (γ_T), all paths will be unacceptable most of the time. Thus SECps will work in the same operation mode as SC to select the best path. On the other hand, if the average SNR per path is high in comparison to the switching threshold, all paths will be acceptable and the receiver will use only one path most of the time. Thus, in the high SNR region, SECPs will work in the same operation mode as SSC.

From figure 3.4, we can see that in SECps system, the diversity gain decreases with the increase of the correlation coefficient, as expected. In additional, the fading effect diminishes, and all average bit error probability from different correlation coefficients $(\rho = 0.1, 0.5 \text{ and } 0.9)$ converge to the same value when SNR increases. Note that the diminishing rate is much higher for the channels with a lower amount of fading (higher *m* parameter). For example, we can see from figure 3.4 that, when *m*=3, the average SNR per path which all average bit error rate from different correlation coefficient converge to

the same value is equal to 25 dB less than the value of average SNR per path at 37 dB for the same performance when m=1.

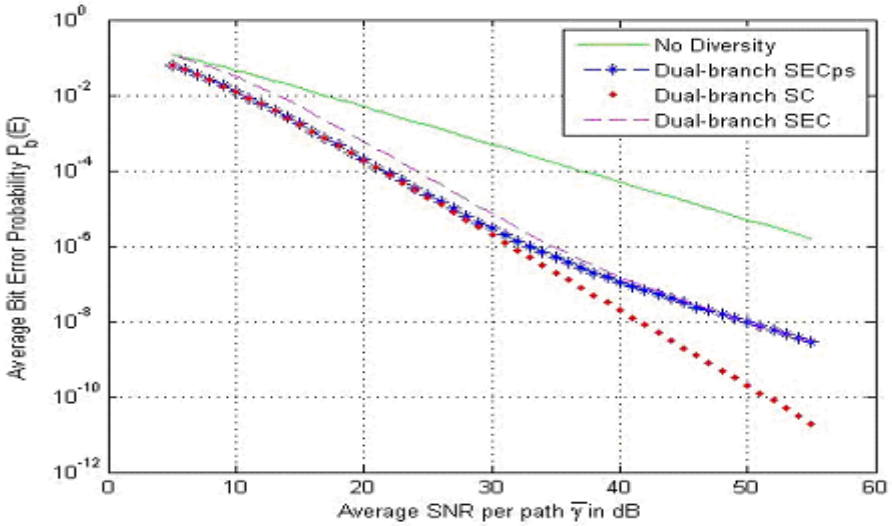


Figure 3.3 Average error rate of BDPSK with dual-branch SECps as function of average SNR $\bar{\gamma}$ over correlated Nakagami-m fading channel in comparison with SC and SSC ($\rho = 0.5, m = 1$ and $\gamma_T = 8$ dB)

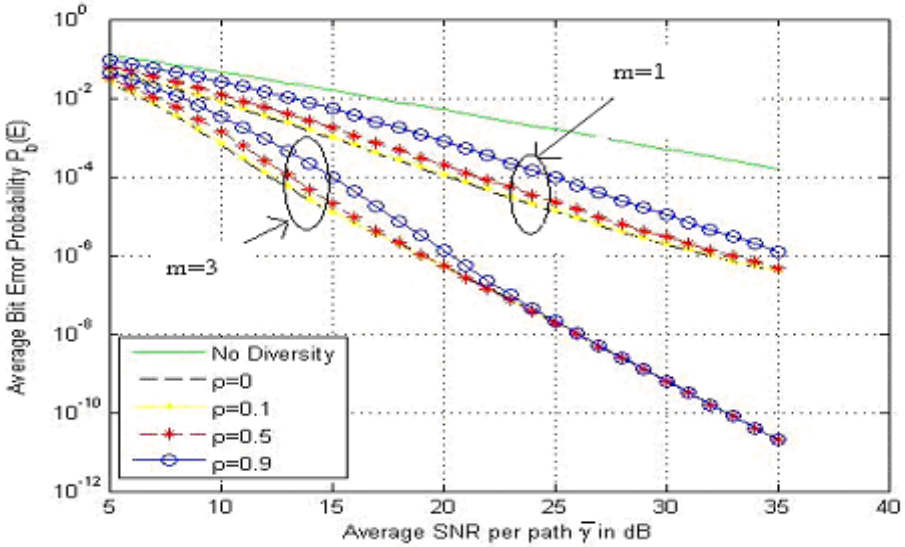


Figure 3.4 Average error rate of BDPSK with dual-branch SECps as function of average SNR $\bar{\gamma}$ over correlated Nakagami-*m* fading channel with various values of correlation coefficient, ρ at different Nakagami parameter,*m*

Chapter 4

Complexity of Dual-Branch SECps

In this chapter, the complexity of dual-branch SECps in term of average number of path estimation and the probability of branch switching are derived and compared with those of dual-branch SSC and dual-branch SC. Comparison of the results for all dual-branch diversity receivers over correlated Nakagami-*m* fading scenario is summarized in table 4.1

4.1 Path Estimation

According to mode of operation of three diversity techniques studied in this report, all combining receivers need to estimate the value of SNR in each diversity path before making a decision. The number of path estimation is thus, depends on the operation of each combining types. For example, the receiver with SC needs to estimate every available diversity path, while the receiver with SSC estimates only necessary path. Since path estimation process requires receiver power consumption, higher number of path estimation results in higher power consumption by the receiver.

In this section, we derive the number of path estimation on dual-branch SECps over correlated Nakagami-*m* fading while the number of path estimation on dual-branch SSC and dual-branch SC are also derived for comparison.

4.1.1 Path Estimation of SECps

Similar to conventional SSC, the SECps receiver estimates diversity paths only if necessary. In additional, the receivers does not need to estimate the other path if the current path is acceptable. Thus, the average number of path estimation denoted by N_E^{SECps} is given in a finite form as follow.

$$N_E^{SECps} = \sum_{i=1}^{2} \pi_i i \tag{4.1}$$

where π_i is the probability that the *i*th diversity paths are estimated. In the case of dual-branch SECps, under identical fading environment, we can see that π_i can be calculated in the probability form as

$$\pi_i = \begin{cases} \Pr[\gamma_T < \gamma_1], & i = 1 \\ \Pr[\gamma_1 < \gamma_T], & i = 2 \end{cases} \tag{4.2}$$

We can rewrite π_i from (4.2) in the term of common CDF, $P\gamma(.)$ and SNR, γ as

$$\pi_i = \begin{cases} 1 - P\gamma(\gamma_T), & i = 1 \\ P\gamma(\gamma_T), & i = 2 \end{cases} \tag{4.3}$$

Therefore, the average number of path estimation on SECps is calculated as follow

$$N_E^{SECps} = \sum_{i=1}^{2} \pi_i i = 1 + P\gamma(\gamma_T) \tag{4.4}$$

On the other hand, conventional SSC receivers need to estimate only one path which is the current path, since when the current path is found unacceptable, the receiver

will switches to the other path whether the SNR value of the switch-to path is acceptable or not. Therefore, the average number of path estimation for SSC is given as

$$N_E^{SSC} = 1 - P\gamma(\gamma_T) \tag{4.5}$$

Since the SNR value of all paths are required in the operation process of SC, the receiver with SC need to estimate all paths. Therefore, in the case of dual-branch SC, the average number of path estimation is given as

$$N_E^{SC} = 2 \tag{4.6}$$

By comparing the average number of path estimation of dual-branch SECps, dual-branch SSC and dual-branch SC in (4.4), (4.5) and (4.6), it can be shown that $N_E^{SSC} \le N_E^{SECps} \le N_E^{SC}$ which means SSC requires the least average number of path estimation, then SECps and SC in respectively.

4.1.2 Simulation and Numerical Results

4.2 Switching Probability

Switching probability in diversity-combining is defined as the probability that a receiver will switch to the other path according to the mode of operation on each type of diversity-combining technique. For instance, the receiver with SC switch to the new path whenever the switch-to path obtains the highest SNR, the receiver with SSC switch to the new path whenever the SNR of the current path falls below the predetermined switching threshold, and the receiver with SECps switches to the new path when the SNR of the current path falls below the threshold and the SNR of the switch-to path is larger than the threshold or that of the current path. Since switching process requires the receiver processing power as well as path estimation process, higher switch probability means more power consumed by the receiver.

In this section, we derive the switching probability of dual-branch SECps over correlated Nakagami-*m* fading while the switching probability of dual-branch SSC and dual-branch SC are also provided for comparison.

4.2.1 Switching Probability of SECps

Based on the mode of operation of dual-branch SECps, we can see that the receiver with SECps switches path only if the SNR of the current path is less than the switching threshold and the SNR of the other path is highest than those of the current path ($\gamma_1 < \gamma_T$ and $\gamma_2 > \gamma_1$ when assuming γ_1 is the current path). Therefore, the switching probability of dual-branch SECps denoted by P_S^{SECps} can be shown as

$$P_S^{SECps} = \Pr[\gamma_1 < \gamma_T \, , \gamma_1 < \gamma_2 \,], \gamma_1, \gamma_2 \geq 0 \tag{4.7}$$

We can write (4.7) in the term of joint PDF as

$$P_S^{SECps} = \int_0^{\gamma_T} \int_{\gamma_1}^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2) d\gamma_2 d\gamma_1 \tag{4.8}$$

From the work of Abu-Dayya and Beaulieu in [6], we can find the close form of $\int_{\gamma_T}^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2) d\gamma$ as follow.

$$\int_{\gamma_1}^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2) d\gamma = \left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1} e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)} Qm\left(\sqrt{\frac{2m\rho\gamma}{(1-\rho)\overline{\gamma}}}, \sqrt{\frac{2m\gamma_1}{(1-\rho)\overline{\gamma}}}\right) \tag{4.9}$$

Therefore, by substituting (4.9) into (4.8), we can get the switching probability as

$$P_S^{SECps} = \int_0^{\gamma_T} \left(\frac{m}{\bar{\gamma}}\right)^m \frac{\gamma_1^{m-1} e^{-(m\gamma_1/\bar{\gamma})}}{\Gamma(m)} Qm\left(\sqrt{\frac{2m\rho\gamma_1}{(1-\rho)\bar{\gamma}}}, \sqrt{\frac{2m\gamma_1}{(1-\rho)\bar{\gamma}}}\right) d\gamma_1 \qquad (4.10)$$

On the other hand, the receiver with dual-branch SSC switches path whenever the current path become unacceptable $(\gamma_i < \gamma_T,\ i \text{ is current path})$. Therefore the switching probability of SSC is given as

$$P_S^{SEC} = \Pr[\gamma_i < \gamma_T,\ i \text{ is current path}] \qquad (4.11)$$

We can write (4.11) in the form of CDF as

$$P_S^{SEC} = P_\gamma(\gamma_T) \qquad (4.12)$$

The switching probability of SSC over correlated Nakagami-*m* fading environment is thus equals the CDF of Nakagami-*m* distributed at $\gamma = \gamma_T$ as follow

$$P_S^{SSC} = P_\gamma(\gamma_T) = 1 - \frac{\Gamma(m, \frac{m}{\bar{\gamma}}\gamma_T)}{\Gamma(m)} \qquad (4.13)$$

In SC case, since all diversity paths are identically faded, it is easy to see that the receiver with dual-branch SC switches path with the probability of $(L-1)/L$ where L is the total number of all diversity paths. Therefore, the switching probability of dual-branch SC is given as

$$P_S^{SC} = (2-1)/2 = 0.5 \qquad (4.14)$$

4.2.2 Simulation and Numerical Results

The complex issue comparison of dual-branch SECps, dual-branch SSC and dual-branch SC over correlated Nakagami-*m* fading channels are given in terms of operation needed for combining decision in the following table.

Table 4.1

Comparison of the combining operation needed over correlated Nakagami-*m* fading channel

Scheme	Average number of path estimations	Probability of branch switching
Dual-branch SC	2	0.5
Dual-branch SSC	1	$P_S^{SSC} = 1 - \frac{\Gamma(m, \frac{m}{\overline{\gamma}}\gamma_T)}{\Gamma(m)}$
Dual-branch SECps	$2 - \frac{\Gamma(m, \frac{m}{\overline{\gamma}}\gamma_T)}{\Gamma(m)}$	$\int_0^{\gamma_T} \left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma_1^{m-1} e^{-(m\gamma_1/\overline{\gamma})}}{\Gamma(m)} Qm\left(\sqrt{\frac{2m\rho\gamma_1}{(1-\rho)\overline{\gamma}}}, \sqrt{\frac{2m\gamma_1}{(1-\rho)\overline{\gamma}}}\right) d\gamma_1$

Chapter 5

Numerical Examples

In this chapter, we illustrate the numerical results of outage probability, average error rate, path estimation and switching probability of dual-branch SECps. The numerical results of dual-branch SSC and dual-branch SC are also given as the comparison.

5.1 Outage Probability

Figure 5.1 shows the outage probability of dual-branch SECps, SSC and SC as the function of normalized outage threshold over correlate Nakagami-m fading environment. While comparing the outage performance of SECps with those of SSC and SC, we can see that SECps system has the same outage probability as SC system when the outage threshold is less than the switching threshold ($\gamma_{th} < \gamma_T$) and as conventional SSC system when the outage threshold is grater than the switching threshold($\gamma_{th} > \gamma_T$). In particular, when $\gamma_{th} < \gamma_T$, the receiver with SECps experiences outage only when the maximum SNR of all diversity paths fall below the outage threshold (γ_{th}) which is the same as the case of the receiver with SC (See appendix B). While when $\gamma_{th} > \gamma_T$, the receiver with SECps experiences outage when the SNR of the first acceptable path falls below the outage threshold (γ_{th}), or when the maximum SNR of all diversity paths is smaller than the switching threshold (γ_T), which is equivalent to the outage performance of conventional SSC. (See appendix C)

In figure 5.2, we study the effect of fading correlation on the outage performance of SECps over Nakagami-*m* fading channel. We use various values of correlation coefficient ($\rho = 0.1, 0.5$ and 0.9) and compare the outage performance over different Nakagami-*m* fading parameters (m=1,3). From this figure, we can see that with fixed normalized outage threshold, the outage probability increases as the fading correlation increases. This is as expected since diversity technique is based on the concept that signals traveling through independent paths are most likely not to experience deep fade simultaneously. Therefore, when the diversity paths are correlated, diversity technique cannot achieve its

optimum performance. When all paths are independent $(\rho = 0)$, the system can achieve the lowest outage probability as we can see from figure5.2.

In additional, it can be seen from figure5.2 that the effect of fading correlation is much more significant in a lower amount of fading (higher *m* parameter) and this effect becomes less significant when $\gamma_{th} > \gamma_T$.

5.2 Average Error Rate

Figure5.33 depicts the average error rate of BDPSK with dual-branch SECps as a function of average SNR per path $(\bar{\gamma})$. The switching threshold is fixed to 8 dB. As we can see, in the low SNR region, dual-branch SECps has the same performance as dual-branch SC, and when SNR increase the performance of SECps degrade and eventually become the same as that of dual-branch SSC. These can be explained as when the average SNR per path $(\bar{\gamma})$ is small in comparison to the switching threshold (γ_T), all paths will be unacceptable most of the time. Thus SECps will work in the same operation mode as SC to select the best path. On the other hand, if the average SNR per path is high in comparison to the switching threshold, all paths will be acceptable and the receiver will use only one path most of the time. Thus, in the high SNR region, SECPs will work in the same operation mode as SSC.

From figure 5.4, we can see that in SECps system, the diversity gain decreases with the increase of the correlation coefficient, as expected. In additional, the fading effect diminishes, and all average bit error probability from different correlation coefficients $(\rho = 0.1, 0.5 \text{ and } 0.9)$ converge to the same value when SNR increases. Note that the diminishing rate is much higher for the channels with a lower amount of fading (higher *m* parameter). For example, we can see from figure 4 that, when *m*=3, all average bit error rate from different correlation coefficient converge to 10^{-8} at average SNR per path = 25 dB less than when m=1 which converge to at average SNR per path = 37 dB.

5.3 Path Estimation

In Figure 5.5(a) and 5.5(b), we study the tradeoff of error performance and path estimations of SECps. Figure 5.5(a) shows the average number of path estimation as a function of the switching threshold (γ_T) for dual-branch SECps and dual-branch SC. Based on the operation mode of SC, the receiver needs to estimate all diversity paths. Therefore, the average number of path estimation for SC always equal to the total number of available diversity paths. For SECps, the average number of path estimation depends on the value of its switching threshold. We can see that the average number of path estimation increases as the switching threshold increases since it is more difficult to find an acceptable path. Therefore, the receiver needs to estimate more and more paths, and eventually the average number of path estimation will reach the maximum point which equal to that of SC when the switching threshold keep increasing. For example, in figure 5.5(a), the average number of path estimation increases from one to its maximum of two paths when the switching threshold increase to 20 dB.

Figure 5.5(b) shows the average bit error probability of dual-branch SECps and dual-branch SC as a function of switching threshold (γ_T). As we can see, the error rate of SECps decreases when the switching threshold increases and becomes eventually the same as that of SC. Note that with the higher correlation coefficient, the smaller switching threshold is required for SECps to provide the same average bit error as that of SC. For example, from table 5.1, it can be shown that 9dB of the switching threshold is required when the correlation coefficient equals 0.1, while the less switching threshold at 7 dB is required when the correlation coefficient equals 0.9.

When consider figure 5.5(a) and 55.(b) jointly, we can see, for example in the case of correlation coefficient equaled 0.5, if the switching threshold is set to 8 dB, SECps can provide the same average bit error as that of SC but needs only 1.3 path estimations while SC need 2 path estimations. Therefore, SECps can achieve the same performance level as those of SC while requiring less path estimation.

Table 5.1

Numerical example of complexity issue requirement on various values of correlation coefficient

Correlation coefficient	Required switching threshold (dB)	Average number of path estimation
0.1	9	1.4
0.5	8	1.3
0.9	7	1.2

5.4 Switching Probability

Figure 5.6 compares the switching probability of dual-branch SECps, dual-branch SC and dual-branch SSC over correlated Nakagami-*m* fading channel. We can see that the switching probability of SECps is the smallest among those of SC and SSC. The switching probability of SECps approaches its maximum of 0.5, which equal to that of SC, when the switching threshold is high in comparison with the average SNR. On the other hand, the switching probability of SSC becomes larger than that of SC and approaches 1, when the switching threshold is high in comparison with the average SNR. Therefore, SECps also saves the receiver operation power by requiring less path switching than conventional SSC and SC.

Figure5.7 compares the switching probability of SECps on various values of correlation coefficients ($\rho = 0.1, 0.5$ and 0.9) and m parameters ($m = 1, 3$) over Nakagami-*m* fading channel. As we can see, while fixing the amount of normalized switching threshold, the switching probability decreases when the correlation coefficient increases. This result is as expected, since in the high correlation condition, a different of SNR in each path is low, thus the receiver can lock to one of the diversity path since no advantage of switching to the other path. However, in the low region of the normalized switching

threshold, the effect of fading correlation diminishes and the switching probability becomes very low. This can be explained as when the switching threshold is much smaller than the average SNR, all paths will be acceptable, thus the receiver will use only one path most of the time. In additional, we can see from figure 5.7 that with fixed normalized switching threshold, the switching probability decreases in a lower amount of fading (higher *m* parameter).

Note, from table 4.1, we can see that the switching probability of SC and SSC do not depend on the value of correlation coefficient.

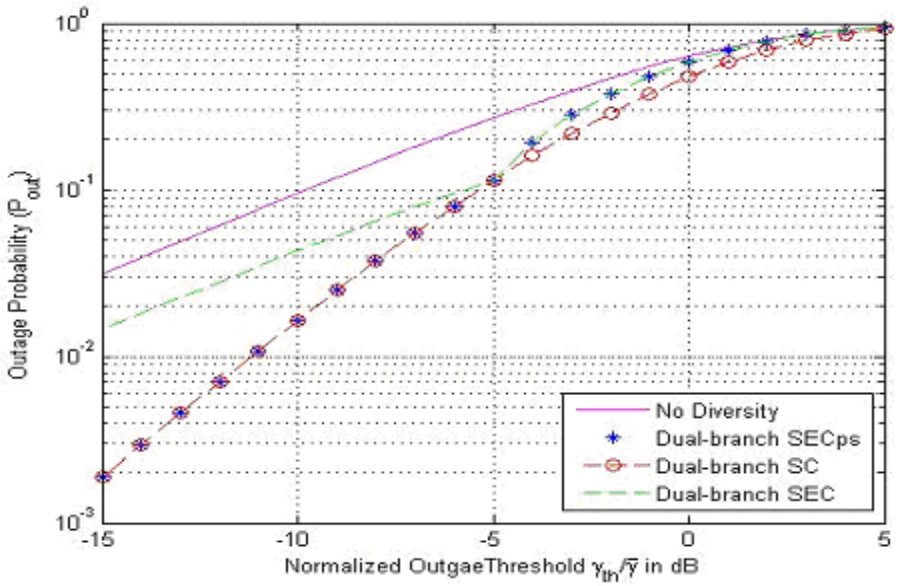


Figure 5.1 Outage probability of dual-branch SECps as function normalized outage threshold $(\gamma_{th}/\overline{\gamma})$ over correlated Nakagami-m fading channel $(\rho = 0.5, m = 1)$ in comparison with SC and SSC $(\gamma_T/\overline{\gamma} = -5$ dB)

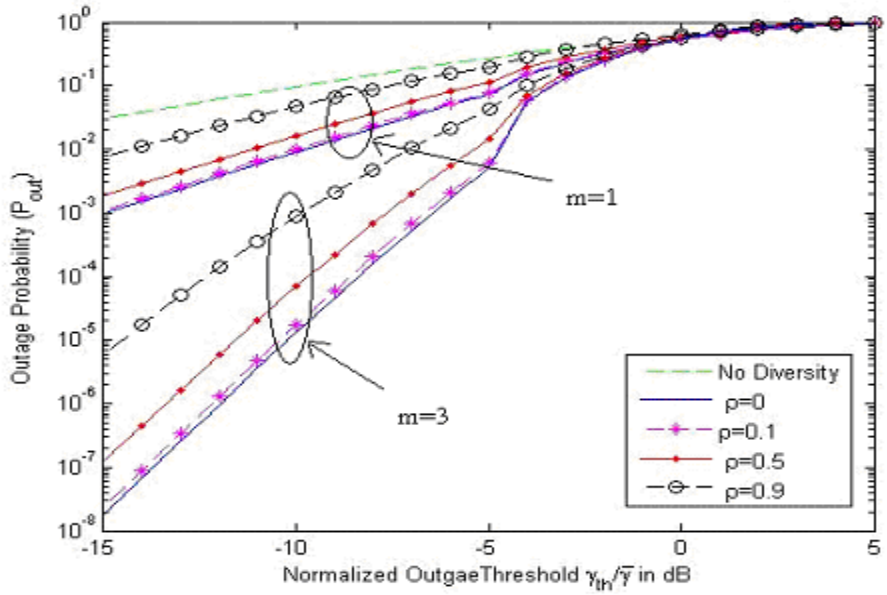


Figure 5.2 Outage probability of dual-branch SECps as function of normalized outage threshold $(\gamma_{th}/\overline{\gamma})$ over correlated Nakagami-*m* fading channel with various values of correlation coefficient $[(a)\rho = 0.1, (b)\rho = 0.5, (c)\rho = 0.9]$ at different *m* parameters [*m=1,3*]

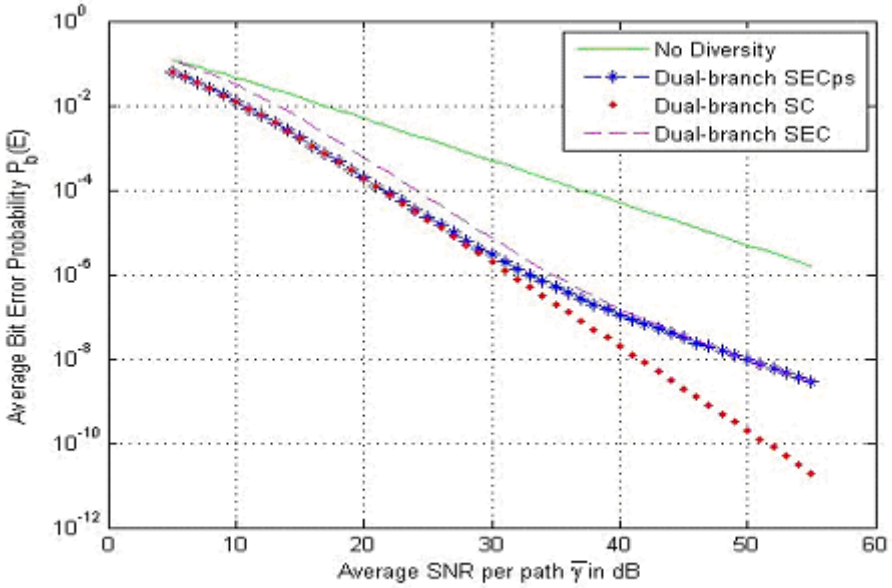


Figure 5.3 Average error rate of BDPSK with dual-branch SECps as function of average SNR $\bar{\gamma}$ over correlated Nakagami-m fading channel $(\rho = 0.5, m = 1)$ in comparison with SC and SSC $(\gamma_T = 8 \text{ dB})$

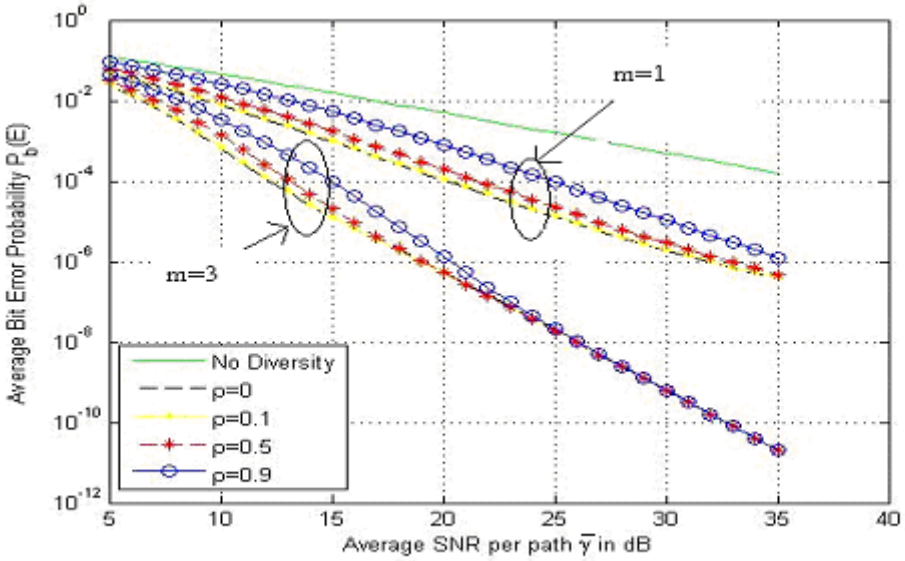


Figure 5.4 Average error rate of BDPSK with dual-branch SECps as function of average SNR $\bar{\gamma}$ over correlated Nakagami-*m* fading channel with various values of correlation coefficient $[(a)\rho = 0.1, (b)\rho = 0.5, (c)\rho = 0.9]$ at different *m* parameters [*m=1,3*]

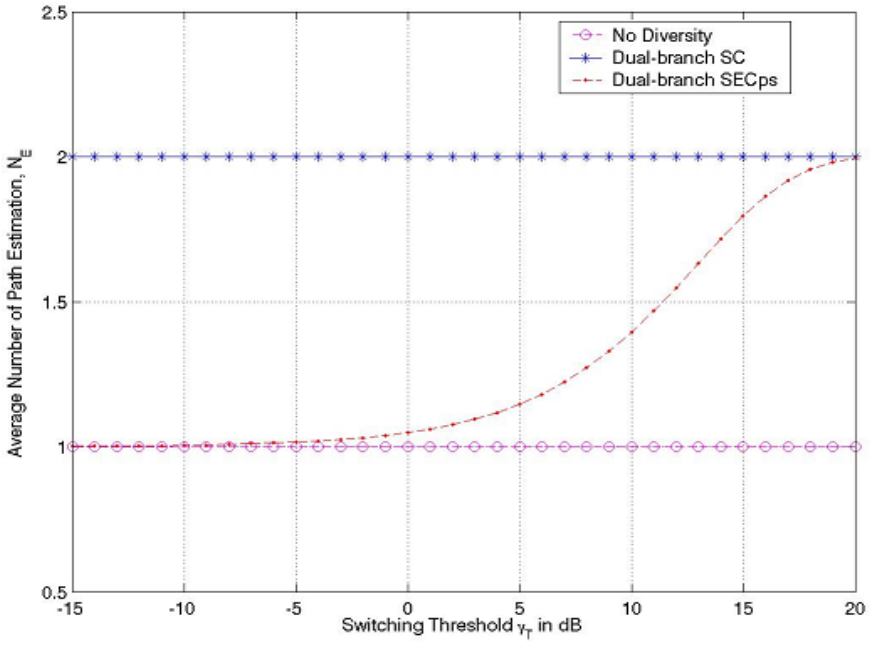


Figure 5.5(a) Trade off error performance versus path estimation (a) Average number of path estimation with SECps

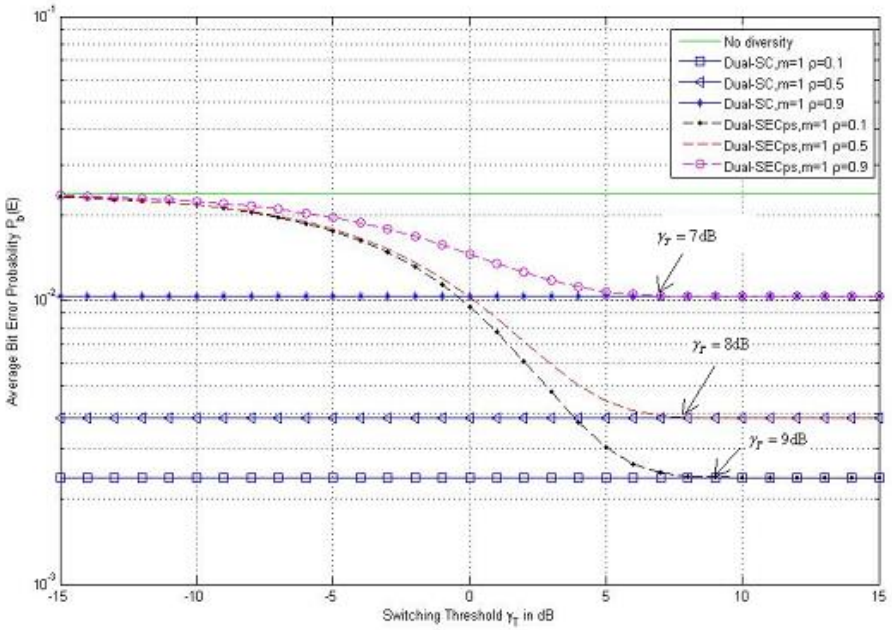


Figure 5.5(b). Trade off error performance versus path estimation (b) Average error rate of DPSK with dual-branch SECps ($\overline{\gamma} = 13$dB)

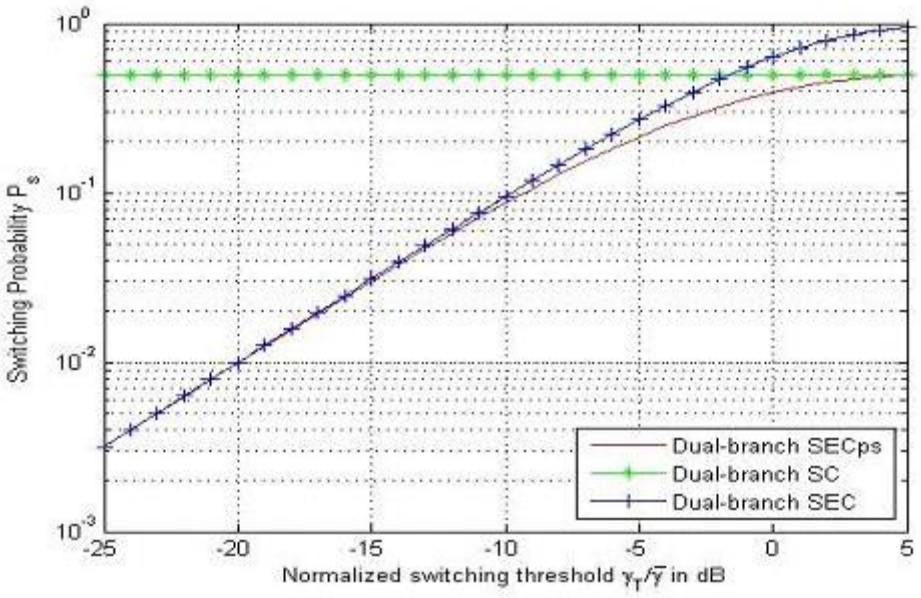


Figure 5.6 Switching probability of dual-branch SECps as function of normalized switching threshold $(\gamma_T / \bar{\gamma})$ over Nakagami-m fading channel (m=1) in comparison with SC and SSC

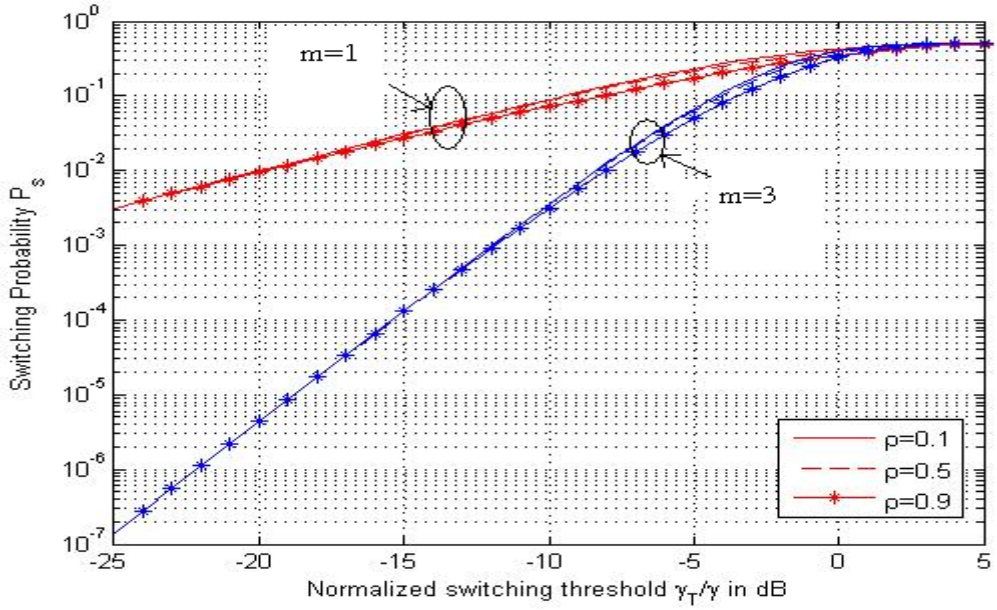


Figure 5.7 Switching probability of dual-branch SECps as function of normalized switching threshold $(\gamma_T / \bar{\gamma})$ over correlated Nakagami-*m* fading channel with various values of correlation coefficient $[(a)\rho = 0.1, (b)\rho = 0.5, (c)\rho = 0.9]$ at different *m* parameters [*m=1,3*]

Chapter6

Conclusion

6.1 Summary and Conclusion

In this report, we investigated the performance of dual-branch SECps over correlated Nakagami-*m* fading environment. We illustrated the improvement in performance of SECps over conventional switch combining scheme. In particular, we performed a tradeoff study of performance versus complexity on SECps. It can be seen that, SECps can achieve much better performance than the conventional SEC at the expense of a little increase in hardware complexity. Also, with the appropriate designed switching threshold, SECps can deliver the same bit error rate as that of SC while requiring less path estimations. In additional, we investigated the effect of fading correlation on the system performance. The numerical result shows that fading correlation induces a non-negligible degradation in the performance of SECps and therefore must be taken into account for the accurate prediction of the performance of SECps.

6.2 Suggestions for Future Work

Since we analyze the performance of SECps only on one condition of identically distributed of correlated fading channels, it is possible to perform the analysis and observe the system performance in different conditions such as

- Analysis of SECps performance over variety of correlation models on fading channel, for example, constant correlation, exponential correlation and arbitrary correlation.
- Analysis of SECps performance over non identically distributed branches.
- Analysis of SECps performance over imperfect channel estimation.

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APPENDIX

A: SSC Output Statistics

The SSC combiner output statistic in appendix A is given by the work of Abu-Dayya and Beaulieu in [6] as follow.

The PDF of SSC is given by [6, eq.(69)] as

$$p_{\gamma_{SSC}}(\gamma)=\begin{cases}A(\gamma) & \gamma<\gamma_T\\ \left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)} + A(\gamma) & \gamma\geq\gamma_T\end{cases} \tag{A.1}$$

where $A(\gamma)$ is defined as

$$\begin{aligned}A(\gamma)&=\int_0^{\gamma_T} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2)d\gamma_2\\ &=\int_0^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2)d\gamma_2-\int_{\gamma_T}^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2)d\gamma_2\end{aligned} \tag{A.2}$$

The first term of (A.2) can be simplified as

$$\int_0^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2)d\gamma=\left(\frac{m}{\overline{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\overline{\gamma})}}{\Gamma(m)} \tag{A.3}$$

While the second term of (A.2) can be expressed in the term of *m*th order Marcum Q-function as

$$\int_{\gamma_1}^{\infty} p_{\gamma_1,\gamma_2}(\gamma_1,\gamma_2)d\gamma = \left(\frac{m}{\bar{\gamma}}\right)^m \frac{\gamma^{m-1}e^{-(m\gamma/\bar{\gamma})}}{\Gamma(m)} Qm\left(\sqrt{\frac{2m\rho\gamma}{(1-\rho)\bar{\gamma}}}, \sqrt{\frac{2m\gamma_1}{(1-\rho)\bar{\gamma}}}\right) \quad \text{(A.4)}$$

B: Outage Probability of SC

For dual-branch SC, the CDF of combiner SNR, γ_Σ is given as

$$P_{\gamma_{SC}}(\gamma) = \Pr[\gamma_\Sigma = \max\{\gamma_1,\gamma_2\} < \gamma] \quad \text{(B.1)}$$

The outage probability of dual-branch SC is defined as the probability that the instantaneous error probability of output SNR falls below a certain specified threshold, γ_{th} as following.

$$P_{out} = \Pr[\gamma_\Sigma = \max\{\gamma_1,\gamma_2\} < \gamma_{th}] \quad \text{(B.2)}$$

C: Outage Probability of SSC

For dual-branch SSC, the CDF of combiner SNR, γ_Σ is given as

$$P_{\gamma_{SSC}}(\gamma) = \begin{cases} \Pr[(\gamma_1 \le \gamma_T)\text{and}(\gamma_2 < \gamma)], & \gamma < \gamma_T \\ \Pr[(\gamma_T \le \gamma_1)\text{or}(\gamma_1 \le \gamma_T \text{and} \gamma_2 < \gamma)], & \gamma \ge \gamma_T \end{cases} \quad \text{(C.1)}$$

The outage probability of dual-branch SC is defined as the probability that the instantaneous error probability of output SNR falls below a certain specified threshold, γ_{th} as following.

$$P_{out} = \begin{cases} \Pr[(\gamma_1 \le \gamma_T) \text{and} (\gamma_2 < \gamma_{th})], & \gamma < \gamma_T \\ \Pr[(\gamma_T \le \gamma_1) \text{or} (\gamma_1 \le \gamma_T \text{and} \gamma_2 < \gamma_{th})], & \gamma \ge \gamma_T \end{cases} \quad \text{(C.2)}$$